

IP₃ contact 1
(IP₃R3 262–274)

ITPR1 human	H	V	F	L	R	T	T	G	R	Q	S	A	T
ITPR2 human	H	I	F	L	R	T	T	L	R	Q	S	A	T
ITPR3 human	Q	V	F	L	R	T	T	L	R	Q	S	A	T
Itpr fly	H	V	F	L	R	T	T	G	R	T	S	A	T
itr-1 worm	P	V	F	L	R	M	T	N	R	P	S	A	A
RYR1 human	D	-	-	-	R	L	V	Y	Y	E	G	G	A
RYR2 human	E	-	-	-	R	T	V	H	Y	E	G	G	A
RYR3 human	S	-	-	-	R	R	I	F	Y	E	A	G	G

IP₃ contact 2
(IP₃R3 499–514)

S	K	P	N	R	E	R	Q	K	L	M	R	E	Q	N	I
T	K	P	N	R	E	R	Q	K	L	M	R	E	Q	N	I
T	K	P	N	R	E	R	Q	K	L	M	R	E	Q	N	I
K	N	P	I	R	D	R	Q	K	L	L	R	E	Q	Y	I
F	S	P	S	R	D	R	Q	K	L	L	R	E	Q	E	V
-	R	S	L	R	N	R	Q	S	L	F	Q	E	E	G	M
-	R	A	L	K	N	R	Q	N	L	F	Q	E	E	G	M
-	R	S	L	K	N	R	Q	N	L	F	K	E	E	G	M

IP₃ contact 3
(IP₃R3 563–573)

S	Q	Q	D	Y	R	K	N	Q	E	Y
S	Q	Q	D	Y	R	K	N	Q	E	Y
S	Q	E	D	Y	R	K	N	Q	E	H
S	Q	Q	D	Y	R	K	N	Q	E	Y
S	Q	V	S	Y	R	K	N	Q	E	F
N	C	A	L	F	S	T	N	L	D	W
N	C	A	Q	F	S	G	S	L	D	W
N	C	A	Q	F	S	N	N	L	D	W

selectivity filter
(IP₃R3 2468–2482)

ITPR1 human	H	G	L	R	S	G	G	G	V	G	D	V	L	R	K
ITPR2 human	Q	G	L	R	N	G	G	G	V	G	D	V	L	R	R
ITPR3 human	H	G	L	R	N	G	G	G	V	G	D	I	L	R	K
Itpr fly	Q	G	L	R	N	G	G	G	I	G	D	I	L	R	A
itr-1 worm	Q	G	L	R	N	G	G	G	I	G	D	V	L	R	N
RYR1 human	V	G	V	R	A	G	G	G	I	G	D	E	I	E	D
RYR2 human	V	G	V	R	A	G	G	G	I	G	D	E	I	E	D
RYR3 human	V	G	V	R	A	G	G	G	I	G	D	E	I	E	D

gate
(IP₃R3 2508–2522)

V	L	N	L	I	F	G	V	I	I	D	T	F	A	D
V	L	N	L	I	F	G	V	I	I	D	T	F	A	D
V	L	N	L	I	F	G	V	I	I	D	T	F	A	D
V	L	N	L	I	F	G	V	I	I	D	T	F	A	D
V	L	N	L	I	F	G	V	I	I	D	T	F	A	D
L	L	A	I	I	Q	G	L	I	I	D	A	F	G	E
L	L	A	I	I	Q	G	L	I	I	D	A	F	G	E
L	L	A	I	I	Q	G	L	I	I	D	A	F	G	E

■ matches the IP₃R consensus ■ differs ■ measured site (IP₃ contact, filter or gate)